

An obstruction theoretic approach to integrated information

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Abstract

Given a physical system, we assign to it a presheaf of local factorization data, whose Čech cohomology measures the failure of local causal factorizations to glue into a compatible global factorization. The first degree obstruction classes for this cohomology theory provide a functorial notion of irreducibility, as suggested by Tull–Kleiner. We prove that vanishing integrated information implies the vanishing of the first degree obstruction classes, showing that they detect the same notion of reducibility as classical Integrated Information Theory. The higher degree obstruction classes naturally define a concept of higher order integration, giving a full hierarchy in a unified framework.

1 Introduction

Integrated Information Theory (IIT) is a mathematical description of the existing consciousness of a system, developed mainly by Tononi and his collaborators, [CITE](#). The theory is centered around the computation of Φ , which is a measure of the *integrated information* in the system. Informally, a system S in a state s , has integrated information, captured by $\Phi(S, s) > 0$, if the information generated by the entire system’s cause-effect structure in state s is greater than the information generated by the combined cause-effect structure of partitions of the system. In other words, it is a measurable emergent property of the system. The formal mathematical framework for computing Φ is rigorous and technical, and uses detailed features of probability theory and measure theory, see [CITE](#).

Classical integrated information is defined through an optimization over partitions and therefore lacks the functorial behavior expected of a categorical invariant. In particular, it is not functorial with respect to subsys-

tem inclusions [CITE](#). Motivated by the work of Tull and Kleiner on a categorical version of IIT [CITE](#), we introduce in the present paper a cohomological formulation of integration, interpreting integration as the obstruction to assembling compatible local causal factorizations into a global factorization. These obstructions live in mathematical objects called *cohomology groups*, which are abstract formulations of the amount of holes a system has in different dimensions. Obstruction theory was originally developed by Stiefel and Whitney to prove the non-existence of certain vector-fields on manifolds [CITE](#), but has later been applied in a plethora of mathematical contexts [CITE](#).

More specifically, given a physical system S in a state s , we assign to every subsystem its local factorization data, organized into a factorization presheaf $\mathcal{F}$. The associated Čech cohomology $H^*(S; \mathcal{F})$ measures whether these local factorizations can be assembled into a compatible global factorization. Intuitively, we wish to build a mathematical structure whose n -dimensional holes contain all the relevant information about n -th order integration. We then define natural first degree obstruction classes $[c^1]$, which we show implies properties for IIT. In particular, if there is a non-trivial obstruction class, then the system has integrated information. These classes detect whether local causal factorizations can be assembled into a compatible global factorization. In this sense, integrated information is interpreted as the failure of descent for causal factorizations.

Theorem 1.1. *Let s be a state of a system S . If there is a non-zero obstruction class $[c^1] \in H^1(S; \mathcal{F})$, then $\Phi(S, s) > 0$.*

The upshot of this work is that one can study integrated information from a new lens, and apply fundamental ideas from algebraic topology and homological algebra to study phenomena in consciousness. In particular, our theory gives a much more general approach to integrated information, that abstracts away many of the optimization-specific features of IIT while retaining

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its notion of reducibility. We expect that our methods can be combined with topological data analysis, see [CITE](#) for an introduction, to locate minimum partitions (MIP) and connect integrated information to new developments in the relationship between topology and information theory, see for example [CITE](#) and [CITE](#).

In the last section we consider a natural extension of our framework: higher order integration. In particular, our framework naturally allows us to introduce higher order obstruction classes $[c^k]$ that detect higher order obstructions to gluing. The above theorem proves that

$$\Phi = 0 \implies [c^1] = 0,$$

but we conjecture that the converse is not true due to the possible non-vanishing of the higher order obstruction classes.

We make no claim surrounding the validity of IIT to explain consciousness and the properties of experience, and make no contributions to the ongoing debates [CITE](#). We only wish to connect some of IIT's mathematical properties to other mathematical concepts known to measure similar features. We view this work not as a new standard for IIT, but as a starting point to explore some of the rich mathematical structures related to it.

2 IIT systems

In [CITE](#), Tull and Kleiner develop a category theoretic foundation for integrated information theory, based on symmetric monoidal categories with some extra structure. We will use this as a backdrop for our cohomology theory, hence we quickly review their setup.

2.1 Process theories

Definition 2.1. A *process theory* is a symmetric monoidal dagger category $(\mathcal{C}, \otimes, \mathbb{1}, \dagger)$, usually denoted simply by $\mathcal{C}$, together with two maps $q_S: S \rightarrow \mathbb{1}$ and $w_S: \mathbb{1} \rightarrow S$ for any $S \in \mathcal{C}$ satisfying monoidality and standard naturality conditions. Objects in $\mathcal{C}$ are interpreted as system types, and morphisms as physical processes. The dagger $\dagger: \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is interpreted as a time-reversal operation.

Morphisms of the form $\mathbb{1} \rightarrow S$ will be referred to as *states*, and we will write $\text{St}(S)$ for the collection

of states of S . Similarly, morphisms $S \rightarrow \mathbb{1}$ will be referred to as *effects* and morphisms $\mathbb{1} \rightarrow \mathbb{1}$ *scalars*. The designated maps q and w are called the *discarding effect* and the *noise state* respectively, and are interpreted as such.

Definition 2.2. A morphism $f: S \rightarrow S'$ is *causal* if it preserves the discarding effect,

$$q_{S'} \circ f = q_S,$$

and similarly *cocausal* if it preserves the noise state.

Remark 2.3. Tull-Kleiner does not assume $\mathcal{C}$ to be a dagger category for their most general setup, but for concretely tying the categorical version to the IIT algorithm of [CITE](#), as we do in [Section 3.5](#), this is required.

Example 2.4. The standard example of a process theory is the category Class of finite sets, where a morphism $S \rightarrow S'$ assigns to an element $s \in S$ an 'unnormalized probability distribution' of elements in S' , which are maps $S \times S' \rightarrow \mathbb{R}^+$. The time reversal operation is given by $f^\dagger(s, s') = f(s', s)$. Composition of two maps $f: S \times S' \rightarrow \mathbb{R}^+$ and $g: S' \times S'' \rightarrow \mathbb{R}^+$, is defined by

$$f \circ g: S \times S'' \rightarrow \mathbb{R}^+ \\ (s, s'') \mapsto \sum_{s' \in S'} f(s, s') g(s', s'').$$

The monoidal structure is the cartesian product, meaning the unit $\mathbb{1}$ is the singleton set. Product of morphisms is given by

$$f \otimes g(s, s'')(s', s''') = f(s, s') g(s'', s''').$$

The discarding process q_S is the unique process with $q_S(s) = 1$ for all $s \in S$, meaning a process is causal whenever it is stochastic. The noise state is the uniform probability distribution with $w_S(s) = \frac{1}{|S|}$.

2.2 Decompositions and subsystems

The following definition will be the backbone of our functorial approach to integration in [Section 3](#)

Definition 2.5. Let $\mathcal{C}$ be a process theory, and S an object in $\mathcal{C}$. A *decomposition* of S is a pair of objects $A, B \in \mathcal{C}$ together with a causal isomorphism $\psi: S \simeq A \otimes B$. We denote decompositions by (A, B, ψ) , or sometimes simply by (A, B) .

Any object $S \in \mathcal{C}$ has a decomposition $(S, \mathbb{1})$, which we call the *trivial* decomposition. We will further say that two decompositions (A, B, ψ) and (A', B', ψ') are *equivalent* if there are causal isomorphisms $f: A \simeq A'$ and $g: B \simeq B'$, such that

$$\psi' = (f \otimes g) \circ \psi.$$

The *complement* decomposition of (A, B) is the same decomposition, just in the opposite order; it is denoted $(A, B)^\perp = (B, A)$. Note that, if two decompositions are equivalent, then so are their complements.

Definition 2.6. Let $\mathbb{D}(S)$ denote the set of equivalence classes of decompositions of S . The operation $(-)^\perp$ acts on $\mathbb{D}(S)$, and we define a *decomposition set* of S to be a subset $\mathbb{D} \subseteq \mathbb{D}(S)$ that contains the trivial decomposition and is closed under $(-)^\perp$.

Definition 2.7. Given a decomposition set $\mathbb{D}$ of S , and a decomposition $(A, B, \psi) \in \mathbb{D}$, we define the *restricted* decomposition set $\mathbb{D}|_A$ to be the set of decompositions of A that can be extended to a decomposition of S in $\mathbb{D}$. Furthermore, we define the *restriction* of a state $s: \mathbb{1} \rightarrow S$ to A to be the state defined by

$$s|_A = (\text{Id}_A \otimes q_B) \circ \psi \circ s: \mathbb{1} \rightarrow A,$$

and similarly for B .

Intuitively, the restriction is defined by identifying S with its decomposition, and then discarding the factor we don't want to consider. By functoriality and naturality, such restrictions depend only on the equivalence class in $\mathbb{D}(S)$, and not on the particular representative.

Definition 2.8. A *system* in $\mathcal{C}$ is a triple $(S, \mathbb{D}, T)$, where $S \in \mathcal{C}$, $\mathbb{D}$ is a decomposition set of S , and $T: S \rightarrow S$ is a causal process, interpreted as a *time evolution*. It is assumed that also $T^\dagger$ is causal.

Example 2.9. The most basic example is the *trivial* system $(\mathbb{1}, \{\mathbb{1} \otimes \mathbb{1}\}, \text{Id}_{\mathbb{1}})$.

A state of a system $(S, \mathbb{D}, T)$ is a state of S . Given such a state s and a decomposition $(A, B) \in \mathbb{D}$, [CITE](#) constructs subsystems of S .

Definition 2.10. Given a system $(S, \mathbb{D}, T)$ in a state s , and a decomposition $(A, B) \in \mathbb{D}$, there is a *subsystem* $(A, \mathbb{D}|_A, T|_A)$, defined by the restricted decomposition

set $\mathbb{D}|_A$ and a restricted time evolution $T|_A$ defined by the composition

$$A \xrightarrow{1 \otimes s|_B} A \otimes B \xrightarrow{T} A \otimes B \xrightarrow{1 \otimes q_A} A.$$

Remark 2.11. These subsystems are again independent of the class representative of the decomposition (A, B) in $\mathbb{D}$.

2.3 Cause and effect structures

Central to IIT is the cause and effect structures of the system S . This is what models how each part of the system can have a causal effect on other parts of the system.

Definition 2.12. A *mechanism-purview pair* is a pair of subsystems of a given system $(S, \mathbb{D}, T)$, depending on the same global state $s: \mathbb{1} \rightarrow S$.

Intuitively a mechanism describes the subset of S that we are “letting make a difference”, and the purview describes the subset of S that is affected.

Definition 2.13. Given a mechanism $(M, \mathbb{D}|_M, T|_M)$ and a purview $(P, \mathbb{D}|_P, T|_P)$, we define the *cause* to be process

$$\text{cau}: M \xrightarrow{1 \otimes w_{M'}} M \otimes M' \simeq S \xrightarrow{T} S \simeq P \otimes P' \xrightarrow{1 \otimes q_{P'}} P,$$

where M' and P' are the complements in the decompositions defined by M and P respectively. Similarly, the *effect* is defined as

$$\text{eff}: M \xrightarrow{1 \otimes w_{M'}} M \otimes M' \simeq S \xrightarrow{T^\dagger} S \simeq P \otimes P' \xrightarrow{1 \otimes q_{P'}} P.$$

Note that both cau and eff send causal states to causal states. Intuitively, these capture the way the previous and next state of P is constrained by the restricted state $s|_M$. When the mechanism is the unit $\mathbb{1}$, then the cause and effect $\text{cau}, \text{eff}: \mathbb{1} \rightarrow P$ are respectively called the *unconstrained cause*, denoted $u\text{cau}$ and the *unconstrained effect*, denoted $u\text{eff}$.

Definition 2.14. Given a state s and a mechanism-purview pair M, P , the associated *cause repertoire* is defined to be the state

$$\text{cau}(s, M, P) = \mathbb{1} \xrightarrow{s|_M} M \xrightarrow{\text{cau} \otimes u\text{cau}} P \otimes P' \simeq S.$$

One defines an *effect repertoire* $\text{eff}(s, M, P)$ similarly.

We also need to decompose and causes and effects.

Definition 2.15. Given a decomposition (M_1, M_2) of a mechanism M , and a decomposition (P_1, P_2) of a purview P , the *decomposed* cause is the process

$$\text{cau}_{M_1, M_2}^{P_1, P_2} : M \simeq M_1 \otimes M_2 \xrightarrow{\text{cau} \otimes \text{cau}} P_1 \otimes P_2 \simeq P,$$

and again similarly for defining the *decomposed* effect.

The *decomposed* cause and effect repertoires are then defined similarly to the normal ones, by replacing cau and eff with their decomposed variants in the definitions. We denote these states by $\text{cau}_{M_1, M_2}^{P_1, P_2}(s, M, P)$ and $\text{eff}_{M_1, M_2}^{P_1, P_2}(s, M, P)$ respectively.

The following definitions will be key in defining our factorization presheaf later.

Definition 2.16. Given a state s of a system $(S, \mathbb{D}, T)$, its *cause-effect structure* is the set

$$R_S = \{\text{cau}(s, M, P), \text{eff}(s, M, P)\}_{(M, P)}$$

over all mechanism-purview pairs. Similarly, we define the cause-effect structure of a subsystem $(A, \mathbb{D}|_A, T|_A)$ for a decomposition (A, B) .

2.4 Directional cuts

To relate our obstruction theoretic framework to classical IIT later, we will need to define cuts of systems.

Given a system $(S, \mathbb{D}, T)$ and a decomposition $(A, B, \psi) \in \mathbb{D}$, one needs to be able to form systems that disconnects certain causal connections in the total system. IIT 3.0 [CITE](#) does this by introducing *cut systems*, that directionally disconnects causal flow from the two parts. IIT 4.0 [CITE](#) does this in a slightly different manner, by replacing parts of the inputs of the time evolutions with independent noise. We have included versions of IIT 4.0's disconnections in arbitrary system states for reference and intuition, but we remark that the actual implementation of such disconnections does not matter much, as long as they are functorial.

Definition 2.17. Given a decomposition $(A, B, \psi) \in \mathbb{D}$, the *left cut* replaces the contribution of B to the time evolution T with the noise state, meaning that only the contribution from A matters. Formally, it is

the system $(S, \mathbb{D}, T^{\psi, \leftarrow})$, where the new time evolution operator is defined as

$$T^{\psi, \leftarrow} : A \otimes B \xrightarrow{1 \otimes w_B q_B} A \otimes B \xrightarrow{T} A \otimes B.$$

In spirit this forces the time evolution to retain all self-influences from A , and influences from A to B , but removes all influences from B to A . The arrow in the notation is meant to denote the direction that is removed.

We define the *right cut* similarly by replacing T with

$$T^{\psi, \rightarrow} : A \otimes B \xrightarrow{w_A q_A \otimes 1} A \otimes B \xrightarrow{T} A \otimes B,$$

and the *double cut* by the diagram

$$\begin{array}{ccc} A \otimes B & \xrightarrow{T^{\psi, \leftrightarrow}} & A \otimes B \\ \downarrow 1 \otimes w_A \otimes w_B \otimes 1 & & \uparrow 1 \otimes q_B \otimes q_A \otimes 1 \\ A \otimes B \otimes A \otimes B & \xrightarrow{T \otimes T} & A \otimes B \otimes A \otimes B \end{array}$$

Remark 2.18. The double cut corresponds to Tull-Kleiner's bidirectional cut, where one evolves the system as if A and B were completely disconnected from each other.

We will use the notation $T^{\psi, \alpha}$, with $\alpha \in \{\leftarrow, \rightarrow, \leftrightarrow\}$ to denote a general disconnection based on a decomposition $\psi : S \simeq A \otimes B$. As we did above with the restricted time evolution, we also get definitions of restricted cuts due to the functoriality of discarding and noise; these are denoted $T|_A^{\psi, \alpha}$.

Given a cut direction α , we can also define cut cause and effect repertoires, $\text{cau}^\alpha(s, M, P)$ and $\text{eff}^\alpha(s, M, P)$, by using the cut time evolution $T^{\psi, \alpha}$. Here ψ is the decomposition associated to the mechanism M . Combining this with the decomposition of cause and effect repertoires, gives *cut-decomposed* cause and effect repertoires.

3 Integrated cohomology

In this section we present the more novel parts of this paper, which is a cohomology theory for IIT systems. We then build an obstruction theory for decomposing a physical system based on this cohomology theory, and show that it detects the existence of integrated information. Lastly we extend this to naturally define higher order integration, often referred to as higher *synergy*.

3.1 Prerequisites

In essence, cohomology groups are mathematical constructions that measure the amount of holes of different dimensions a space or other mathematical object has. It is generally defined using homological algebra, see [CITE](#) for an introduction.

Definition 3.1. A *cochain complex* $C^\bullet$ is a sequence of abelian groups

$$\dots \longrightarrow C^{i-1} \xrightarrow{\delta^{i-1}} C^i \xrightarrow{\delta^i} C^{i+1} \longrightarrow \dots$$

such that $\delta^{i-1} \circ \delta^i = 0$ for all i .

The elements of $C^\bullet$ are called the *i-cochains*. An *i-cochain* in the image of the map δ^{i-1} is called a *coboundary*, while an *i-cochain* in the kernel of δ^i is called a *cocycle*. The equation $\delta^{i-1} \circ \delta^i = 0$ implies that $\text{Im}(\delta^{i-1}) \subseteq \text{Ker}(\delta^i)$, which is the fundamental fact used to define cohomology – in words, any coboundary is also a cocycle.

Definition 3.2. Given a cochain complex $C^\bullet$, its i 'th *cohomology group* is defined as the quotient group

$$H^i(C^\bullet) := \text{Ker}(\delta^i) / \text{Im}(\delta^{i-1}).$$

Elements of $H^i(C^\bullet)$ are called *cohomology classes*.

One of the central goals of algebraic topology and algebraic geometry is to functorially assign cochain complexes to topological and geometric objects and compute the resulting homology groups. For a category of interest $\mathcal{C}$, for example topological spaces, manifolds or schemes, such assignments defines functors into the category of graded abelian groups $H^*: \mathcal{C} \longrightarrow \text{Ab}_*$ called *cohomology theories*. Examples include *simplicial cohomology* [CITE](#), *de Rham cohomology* [CITE](#), *coherent cohomology* [CITE](#) and Čech cohomology. It is the latter one we use in this paper, so let us quickly recall its construction. The reader is referred to [CITE](#) for details.

Definition 3.3. Čech cohomology

3.2 The factorization presheaf

Definition 3.4. Let $(S, \mathbb{D}, T)$ be a system in a state s . We define the repertoire presheaf $\mathcal{R}$ to be the functor

$$\begin{aligned} \mathcal{R}: \text{Sub}_s(S)^{\text{op}} &\longrightarrow \text{Ab} \\ U &\longmapsto \mathbb{Z}[R_U] \end{aligned}$$

mapping a subsystem U to the free group on equivalence classes of cause-effect structures on U . Given a subsystem inclusion $V \hookrightarrow U$, the restriction map $\rho_U^V: \mathcal{R}(U) \longrightarrow \mathcal{R}(V)$ is defined as the equivalence class $[R_{U|V}]$, extended linearly.

Definition 3.5. Let r be a repertoire (either a cause or effect) and (A, B) a decomposition of S . The restrictions

$$r|_A: \mathbb{1} \longrightarrow A \text{ and } r|_B: \mathbb{1} \longrightarrow B$$

define a new state

$$r|_A \otimes r|_B: \mathbb{1} \simeq \mathbb{1} \otimes \mathbb{1} \longrightarrow A \otimes B \simeq S$$

which we call the *factorized repertoire*. We say r *factors* over the decomposition (A, B) if $r = r|_A \otimes r|_B$, and we say that r is *factorizable* if it factors over some decomposition of S .

3.3 Integrated cohomology

Given a system $(S, \mathbb{D}, T)$ and a global state $s: \mathbb{1} \longrightarrow S$, we will denote by $\text{Sub}_s(S)$ the category of subsystems of S , defined as the poset category associated to the poset structure in [CITE](#). In other words, $\text{Sub}_s(S)$ is the category consisting of subsystems as objects, and morphisms being single comparison maps by the pre-order structure.

The overall idea of our approach can be described as follows:

1. first, define a geometric object $\mathcal{U}$ based on our system S , which in this case will be a Čech-cover of $\text{Sub}_s(S)$;
2. second, define a presheaf $\mathcal{F}$ of on $\text{Sub}_s(S)$ that describes the cause-effect factorizations of S ;
3. third, define the integrated cohomology of S to be the Čech cohomology valued in this presheaf;
4. and lastly, prove that the failure of descent for the Čech cover detects integrated information in S .

Definition 3.6. Čech cover of $\text{Sub}_s(S)$

There are probably several possible choices for what presheaf $\mathcal{F}$ to use, depending on how categorical or how system specific one wants to be. If one focuses on actual IIT 4.0 constructions, then perhaps a probability based approach using intrinsic information and

intrinsic difference would be a good strategy. We have, however, opted for a more categorical and general approach. This is in part to give generalized integrated information functorial properties, as discussed by Tull–Kleiner in [CITE](#).

Definition 3.7. Let r be a repertoire (either a cause or effect) and (ψ, A, B) a decomposition of S . The restrictions

$$r|_A: \mathbb{1} \longrightarrow A \text{ and } r|_B: \mathbb{1} \longrightarrow B$$

define a new state

$$r^\psi := r|_A \otimes r|_B: \mathbb{1} \simeq 1 \otimes 1 \longrightarrow A \otimes B \simeq S$$

which we call the *factorized repertoire*. We say r *factors* over the decomposition (A, B) if $r = r^\psi$, and we say that r is *factorizable* if it factors over some decomposition of S .

Definition 3.8. Given a subsystem $U \in \text{Sub}_s(S)$, its *local factorization data* $\text{Fact}(U)$ is the set of equivalence classes of pairs (ψ_U, r_U) , where ψ_U is a decomposition of U and r_U is a repertoire that factors over ψ , in other words $r_U = r_U^\psi$.

Given a subsystem $V \subseteq U$ and a decomposition (ψ_U, A, B) of U , we can define an induced decomposition on V by restricting the factors A and B through the chosen decomposition of V . Using this we obtain the following lemma.

Lemma 3.9. *Let $\psi_U = (A, B) \in \mathbb{D}|_U$ and $r_U = r_U|_A \otimes r_U|_B$ be a local factorization datum on a subsystem U . If $V \subseteq U$ is a subsystem of U , then the restricted repertoire $r_U|_V$ factors over the induced decomposition $\psi_U|_V$.*

Proof. Let the induced decomposition of V be $\psi_U|_V = (A_V, B_V)$, where A_V and B_V are obtained by restricting the factors A and B along the chosen decomposition of V .

Using transitivity of restriction,

$$(r_U)|_{A_V} = (r_U|_A)|_{A_V}$$

and similarly for B . Since restriction is defined by discarding complementary factors, naturality and monoidality of the discarding maps implies that

$$(r_U|_A \otimes r_U|_B)|_V = (r_U|_A)|_{A_V} \otimes (r_U|_B)|_{B_V}.$$

Hence, $r_U|_V = (r_U)|_V = (r_U)|_{A_V} \otimes (r_U)|_{B_V}$, showing that $r_U|_V$ factors over the induced decomposition $\psi_U|_V$. $\square$

Definition 3.10. The *factorization presheaf* is defined as the functor

$$\mathcal{F}: \text{Sub}_s(S)^{\text{op}} \longrightarrow \text{Ab}$$

sending a subsystem U to the free group on $\text{Fact}(U)$. An inclusion $V \subseteq U$ defines a by [Lemma 3.9](#) a restriction map $\rho_V^U: \mathcal{F}(U) \longrightarrow \mathcal{F}(V)$.

The factorization presheaf is not always a sheaf with respect to the decomposition topology on $\text{Sub}_s(S)$. In fact, our claim is that its failure to be a sheaf is detected by integrated information in S .

Remark 3.11. Our setup is formally analogous to Abramsky–Brandenburger’s sheaf theoretic description of contextuality, see [CITE](#) and [CITE](#). The similarities are enticing, and we feel that that the possible interactions are worth further investigation.

Definition 3.12. Let $(S, \mathbb{D}, T)$ be a system in a state s . We define its *integrated cohomology* to be $H_{\text{int}}^*(S) = \tilde{H}^*(\mathcal{U}; \mathcal{F})$, where $\mathcal{U}$ is the decomposition cover of $\text{Sub}_s(S)$ and $\mathcal{F}$ is the factorization presheaf.

The intuition is as follows: The Čech nerve of $\text{Sub}_s(S)$ gives us a geometric construction of *where* integrated information can be tested. It consists formally of chains of substates of S of all lengths, with topological structure that understands how these chains are related and connected. The presheaf gives us *what* is measured on every level. The cohomology gives us information about which local factorizations that can be glued together to form a global factorization. If gluing fails, then there is some level of the system where its global structure cannot be described in terms of local ones, which is the essence of the system having integrated information. In particular, a 1-cocycle measures the failure of a pairwise compatability.

Definition 3.13.

3.4 Generalized IIT-algorithms

In order to prove the main theorem of this paper, we need to be able to compute and understand Φ . We will use the *IIT-algorithm* as developed by Tull–Kleiner in

CITE . This ensures that our theory works not only for classical IIT, but also quantum IIT and other variants that one would like to study, **CITE** .

The IIT-algorithm defines integration for all cause and effect repertoirs, and uses this to aggregate upwards to an integration value for the total system S . This is done by assuming that there is a distance function on the states of S , $d(-, -): \text{St}(S) \times \text{St}(S) \rightarrow \mathbb{R}_{\geq 0}$. As in **CITE** , we do not require this to be a metric, though it usually will be in examples.

Remark 3.14. In classical IIT, this metric $d(-, -)$ is the Wasserstein metric, while in quantum-classical IIT it is the metric induced by the norm of the system C^* -algebra.

Definition 3.15. Let $\text{cau}(s, M, P)$ be a cause repertoire. Its *integration* $\varphi(\text{cau}(s, M, P))$ is defined as the minimum distance from $\text{cau}(s, M, P)$ to a decomposed repertoire associated to a non-trivial decomposition:

$$\min_{(M_1, M_2), (P_1, P_2)} d(\text{cau}(s, M, P), \text{cau}_{M_1, M_2}^{P_1, P_2}(s, M, P)).$$

We define the integration level of an effect repertoire $\text{eff}(s, M, P)$ similarly.

Definition 3.16. Given a state $s: \mathbb{1} \rightarrow S$ and a sub-system $M \in \text{Sub}_s(S)$, the *core cause* is the sub-system $P^{cc} \in \text{Sub}_s(S)$ such that the associated cause repertoire $\text{cau}(s, M, P)$ has maximal integration. Similarly, the *core effect* is the sub-system P^{ce} that maximizes the integration of the associated effect repertoire $\text{eff}(s, M, P)$.

Definition 3.17. The integration of a mechanism M is then defined as

$$\varphi(M) := \min(\varphi(\text{cau}(s, M, P^{cc})), \varphi(\text{eff}(s, M, P^{ce}))).$$

Given a sub-object $M \in \text{Sub}_s(S)$, the triple consisting of the core cause repertoire, the core effect repertoire and the resulting integration defines the *concept* of M . Allowing M to vary, defines the *Q-shape* of the state s . The collection of all *Q-shapes* defines the *experience space* $\mathbb{E}(S)$ of the system S . See **CITE** for the details.

Remark 3.18. One of the fundamental beliefs in IIT is that any conscious experience a system S can have is an element of $\mathbb{E}(S)$.

By minimizing over all possible decompositions, one can similarly define the integration of a *Q-shape*. The

subsystem with maximally integrated *Q-shape* is the system's *major complex*, which by definition defines an element $\mathbb{E}(S, s)$ in the experience space $\mathbb{E}(S)$. IIT claims that this element is the particular conscious experience of the system S in the state s , and the associated Φ -value is defined as the size of this conscious experience: $\Phi(S, s) = \|\mathbb{E}(S, s)\|$.

If this Φ -value is zero, then it means that for all mechanisms M , the integration $\varphi(M)$ is zero, meaning that for all purviews P there is a decomposition with the same integration of its cause and effect repertoire. In principle, decomposing the system does not change its cause and effect repertoirs. We now use this to prove our main result.

3.5 Obstruction classes and generalized IIT

The goal of this section is to prove the following theorem.

Theorem 3.19. *Let s be a state of a system $(S, \mathbb{D}, T)$. If there is no integrated information in S , then all of the decomposition obstruction classes vanish. In other words,*

$$\Phi(S, s) = 0 \implies [\Delta^{\psi, \alpha}] = 0$$

for all A, ψ and α . In particular, the existence of non-vanishing obstruction classes implies $\Phi(S, s) > 0$, and hence that the classes $[\Delta^{\psi, \alpha}]$ form obstructions to the decomposition of causal structures.

4 Extensions

Our framework naturally allows for some extensions and future directions, of which we here outline two. As mentioned before, our framework in this paper only uses the binary value of Φ – it is either zero or non-zero – but the true strength of IIT lies in its *size*. Hence our first extension constructs a proper analogue to the Φ -value in our framework, which we denote Φ_{coh} . Further, our cohomological Φ -value, as well as our previously defined obstruction classes, all lie in the first cohomology groups $H_{int}^1(S)$ of the system $(S, \mathbb{D}, T)$. Utilizing the higher order cohomology groups as well, we obtain higher order obstruction classes and higher order cohomological Φ -values, which are analogous to the higher order synergy and contextuality of **CITE** .

4.1 Cohomological integrated information

In [Section 3.5](#) we introduced a distance function between states,

$$d: \text{St}(S) \times \text{St}(S) \longrightarrow \mathbb{R}_{\geq 0},$$

that we used to describe Tull–Kleiner’s generalized IIT-algorithm. We here use this to infer a distance function on the cohomology groups, which we then use to define the cohomological integrated information Φ_{coh} .

Definition 4.1. Let $A \in \text{Sub}_s(S)$ be a subsystem. The *distance* between two endomorphisms f, g of A is defined by how differently they act on the local state $s|_A$. In other words,

$$\|f - g\|_A := d(f \circ s|_A, g \circ s|_A).$$

This extends linearly to a distance function on $\mathcal{F}(A)$. Furthermore, given a k -cochain $c \in C^k(S; \mathcal{F})$, we define its *size* as its largest value on the set of all k -simplices:

$$\|c\| := \max_{\sigma} \|c(\sigma)\|.$$

In order to infer a size on cohomology classes, we simply need to incorporate the fact that cohomology classes are only defined up to a coboundary, as they are quotient classes of abelian groups. Hence, given a cohomology class $[c] \in H_{int}^k(S)$, we define

$$\|[c]\| := \min_{\theta \in C^{k-1}(S; \mathcal{F})} \|c + \delta\theta\|.$$

This definition makes the value intrinsic, as it minimizes over all possible representations of the class.

Definition 4.2. Let S be a system in a state s . The *cohomological integrated information* is defined as the size of the smallest obstruction class,

$$\Phi_{coh}(S, s) := \min_{\psi, (\alpha_k)} \|\Delta^{\psi, (\alpha_k)}\|.$$

The following result immediately follows from the definition combined with [Theorem 3.19](#).

Proposition 4.3. *If s is a state of a system $(S, \mathbb{D}, T)$ such that $\Phi(S, s) = 0$, then also $\Phi_{coh}(S, s) = 0$.*

We hope to revisit this definition in future work and connect it to the definition of the standard integrated information value $\Phi(S, s)$. For now we explain how this

solves Tull and Kleiner’s search for a functorial construction of integrated information.

In order to have functoriality, we need to understand what categories are present. In particular, we need a category that describes both systems $(S, \mathbb{D}, T)$ and their states s . The recipient category will be the poset category $\mathbb{R}_{\geq 0}$ of non-negative real numbers. A map into this poset category is functorial when it is monotone.

Definition 4.4. A *system state* in a process theory $\mathcal{C}$, is a tuple $(S, \mathbb{D}, T, s)$, where $(S, \mathbb{D}, T)$ is a system in $\mathcal{C}$ and $s: \mathbb{1} \longrightarrow S$ is a state of S . A *morphism* of system states is a causal map

$$(S, \mathbb{D}, T, s) \longrightarrow (S', \mathbb{D}', T', s')$$

that

1. preserves states: $f \circ s = s'$;
2. preserves dynamics: $f \circ T = T'$;
3. preserves decompositions: for a decomposition (A, B, ψ) there exists a decomposition $(f(A), f(B), \psi')$ into subsystems identified by f ;
4. induces a functor on the poset categories: $f_*: \text{Sub}_s(S) \longrightarrow \text{Sub}_{s'}(S')$;
5. is compatible with the restriction maps: $f(\rho_U^V(c)) = \rho_{f(U)}^{f(V)(f(c))}$;
6. is non-expansive: $\|f_*(c)\| \leq \|c\|$.

The resulting category of system states and morphisms is denoted $\text{Sys}(\mathcal{C})$.

It follows from the definition of the category above that the cohomological integrated information is a functor, as the obstruction classes are functorial by design, and the properties of morphisms in $\text{Sys}(\mathcal{C})$ ensures that it is a monotone map. Hence we have the following proposition.

Proposition 4.5. *Let $\mathcal{C}$ be a process theory. The cohomological integrated information is a functor*

$$\Phi_{coh}: \text{Sys}(\mathcal{C}) \longrightarrow \mathbb{R}_{\geq 0}.$$

In particular, isomorphic system states have equal cohomological integrated information,

4.2 Higher order integration

Our setup of using obstruction theory easily allows us to extend integrated information to higher orders – something that cohomology theory is a natural arena for. There are some attempts at this already, see for example [CITE](#), but we will save comparisons for future work.

Higher obstruction classes are defined using more refined decomposition of the system S . The intuition is that, given a decomposition $\psi: S \simeq A \otimes B \otimes C$, the Φ -values computed pairwise from these could all be zero, even if the total Φ -value is not. This failure should be measured not by the bipartite obstruction classes previously defined, but by a higher obstruction class.

Definition 4.6. Let S be an object in a process theory $\mathcal{C}$. A k -ary decomposition of S is a k -tuple $(A_1, A_2, \dots, A_k)$ together with a causal isomorphism

$$\psi: S \simeq A_1 \otimes A_2 \otimes \dots \otimes A_k.$$

We will denote such a decomposition simply by (A_k) . In particular, a k -ary decomposition is an iterated binary composition, hence naturally lives in $\mathbb{D}(S)$.

By inserting noise, we can use these to construct *multi-cut* time evolutions $T^{\psi, (\alpha_k)}$, where (α_k) is a sequence of k directions, also allowing for some of the values being empty.

Definition 4.7. Given a system $(S, \mathbb{D}, T)$, a k -ary decomposition (A_k) and a cut-vector (α_k) , we define the associated obstruction class on a $k+1$ -simplex $\sigma = U_0 \rightarrow \dots \rightarrow U_k$ as the cohomology class of

$$\Delta^{\psi, (\alpha_k)}(\sigma) = \rho_{U_0}^{U_k}(T|_{U_k}) - \rho_{U_0}^{U_k}(T^{\psi, (\alpha_k)}|_{U_k}),$$

where $\rho_{U_0}^{U_k}$ is the iterated restriction from U_k to U_0 defined by σ .

This is a class living in $H_{int}^k(S)$, which measures the failure of the largest subsystem U_k to decompose into k parts that are “ (α_k) -independent” – if the direction vector consisted of only bidirectional arrows, then these parts would have been independent. For the case $k=1$, these classes reduce to the ones studied earlier.

Analogously to the construction above for $\Phi_{coh}(S, s)$ we can define higher order cohomological integrated information.

Definition 4.8. Let $(S, \mathbb{D}, T, s)$ be a system state. Its k -th order cohomological integrated information is defined as the size of the smallest higher order obstruction class:

$$\Phi_{coh}^k(S, s) := \min_{\psi, (\alpha_k)} \|\Delta^{\psi, (\alpha_k)}\|.$$

As in [Proposition 4.5](#), these higher order classes are functorial, and for $k=1$ we have $\Phi_{coh}^1(S, s) = \Phi_{coh}(S, s)$ as defined above.

This defines for a system state $(S, \mathbb{D}, T, s)$ a sequence of real numbers $\Phi_{coh}^1(S, s), \Phi_{coh}^2(S, s), \Phi_{coh}^3(S, s), \dots$ that detect higher and higher orders of integration. We round off the paper with the following definition and conjecture.

Definition 4.9. The integration order of a system state $(S, \mathbb{D}, T, s)$ is defined as the smallest k such that $\Phi_{coh}^k(S, s) \neq 0$. If all higher cohomological integrated information vanishes, i.e., $\Phi_{coh}^k(S, s) = 0$ for all k , then we say it has integration order ∞ .

Conjecture 4.10. If $(S, \mathbb{D}, T, s)$ is a system state with integration order ∞ , then $\Phi(S, s) = 0$.